Personal Digital Twin

A Data Engineering Platform for Personal Behavior Analytics and What-if Simulation

Project Type: Data Engineering and Analytics Project

Project Stage: Project Proposal

Prepared by:

Ahmed Mohamed Khalaf
Ahmed Mohamed Shehata
Badr Mostafa
John Sami
Mariam Essam
Ayat Khalaf

1. Abstract

The Personal Digital Twin is a data engineering project designed to create a digital representation of an individual's daily habits, activities, and behavioral patterns. The system collects personal lifestyle data, including sleep duration, study hours, physical activity, screen time, task completion, and daily productivity. It transforms this data into structured, meaningful information that can be used for analysis, visualization, and exploratory prediction.

The main purpose of the project is to demonstrate how data engineering techniques can transform raw daily records into a reliable analytical system. The proposed platform will implement a data pipeline that includes data collection, data cleaning, transformation, storage, and analytical reporting.

The system will organize historical data in a structured database and present important indicators through an interactive dashboard. Users will be able to explore patterns over time, compare different periods, identify relationships between habits and productivity, and simulate hypothetical changes to their daily routines.

For example, a user may compare their current routine with a hypothetical routine that includes an additional hour of sleep or a different study schedule. The platform will display

the available historical evidence and estimated changes based on the selected scenario, while clearly distinguishing estimates from guaranteed outcomes.

The project will focus on data integration, ETL processes, data modeling, data quality, analytical queries, visualization, and the foundations of predictive analytics. It will use sample or user-provided data and will not require access to private device information.

Keywords: Personal Digital Twin, Data Engineering, ETL Pipeline, Data Warehouse, Behavioral Analytics, Data Visualization, Predictive Analytics, What-if Analysis.

2. Introduction

People generate different types of personal data every day through their routines and activities. Examples include the number of hours they sleep, the time they spend studying, their physical activity, their screen time, and the tasks they complete.

Although these records may appear simple, analyzing them together can reveal patterns that are difficult to notice through manual observation. For example, a person may want to understand whether their study time is consistent throughout the week, how their sleep schedule changes during examination periods, or whether their daily productivity differs between weekdays and weekends.

However, personal information is often stored in separate applications, spreadsheets, notebooks, or devices. This makes it difficult to combine the information into one reliable and comprehensive view.

The Personal Digital Twin project addresses this challenge by creating a unified data platform that integrates personal activity records and transforms them into useful analytical information.

In this project, a digital twin refers to a data-driven digital representation of an individual's recorded habits and activities. It is not intended to be a complete biological replica of a human being. Instead, it is a structured analytical model that reflects selected aspects of a person's routine based on available data.

The platform will demonstrate how data engineering can support personal analytics by building a complete workflow from raw data to actionable insights.

3. Problem Statement

Personal lifestyle data is often fragmented across multiple sources and recorded in different formats. As a result, users may struggle to understand long-term behavioral patterns and the relationships between their daily activities.

Traditional habit-tracking methods usually focus on recording individual metrics. For example, one application may track sleep, while another records exercise or study time. These tools do not always provide an integrated analytical view of the user's activities.

Furthermore, raw data may contain missing values, inconsistent formats, duplicate records, or incorrect entries. Without proper data processing, the resulting analysis may be incomplete or misleading.

The main problem addressed by this project is the absence of a unified and structured data engineering pipeline that transforms multiple categories of personal activity data into a consistent analytical model.

The proposed system will address this problem by integrating data from selected sources, cleaning and transforming records, storing the processed data in a structured database, and using analytical techniques to identify patterns and support what-if comparisons.

The project will also investigate how historical data can be used to produce exploratory estimates, while acknowledging that observed relationships do not necessarily establish causation.

4. Project Aim

The main aim of the Personal Digital Twin project is to design and implement a data engineering platform that creates a data-driven representation of an individual's daily habits and supports personal behavior analysis through integrated data processing, structured storage, visualization, and scenario-based analysis.

The project will demonstrate the complete data lifecycle, beginning with raw data collection and ending with analytical outputs that help users understand their recorded routines.

5. Project Objectives

The project has the following objectives:

1. To collect personal activity data from structured CSV files, manual entries, or other permitted data sources.

2. To integrate multiple categories of personal data into a unified data model.

3. To design an ETL pipeline that extracts, cleans, transforms, validates, and loads the collected records.

4. To handle missing values, duplicate records, inconsistent formats, and invalid entries.

5. To design and implement a relational database for storing personal activity records and analytical information.

6. To organize historical data in a form suitable for reporting and time-based analysis.

7. To calculate meaningful metrics related to sleep, study, physical activity, screen time, and task completion.

8. To develop an interactive dashboard for visualizing personal activity patterns.

9. To identify trends and relationships between different recorded habits.

10. To implement a basic what-if analysis module that compares the observed routine with hypothetical changes.

11. To explore simple forecasting techniques when sufficient historical data is available.

12. To evaluate the system in terms of data quality, pipeline reliability, analytical usefulness, and performance.

13. To apply privacy-conscious practices when collecting, storing, and processing personal information.

6. Proposed Solution

The proposed solution is a centralized personal analytics platform built around a data engineering pipeline.

The platform will accept activity records from supported input sources, standardize the records, validate their quality, and store them in a structured database.

Once the data has been processed, the analytical layer will calculate daily and weekly indicators. These indicators will be displayed through an interactive dashboard that allows users to explore changes over time and compare different activity categories.

A separate what-if analysis component will allow users to adjust selected hypothetical inputs, such as planned study hours or sleep duration, and compare the scenario with their historical records.

For the initial version, scenario results will be based on transparent rules, historical averages, or simple statistical models. More advanced machine learning techniques may be added only if the available data is sufficient to justify their use.

The platform will not claim to predict a person's behavior with certainty. Instead, it will provide evidence-based summaries and clearly labeled estimates that help users explore possible outcomes.

7. Main Features

7.1. Personal Data Collection

The system will collect selected information about the user's daily routine.

The initial data categories will include:

- Sleep duration.

- Study duration.

- Physical activity duration.

- Screen time.

- Number of completed tasks.

- Total planned tasks.

- Daily self-reported productivity score.

- Date and day of the week.

Data may be entered manually or imported from CSV files. The system will not depend on paid services, wearable devices, or private accounts for its initial implementation.

7.2. Data Cleaning and Validation

The collected records will pass through a data quality stage before they are stored for analysis.

This stage will identify duplicate records, invalid numerical values, missing fields, inconsistent date formats, and values outside reasonable configured ranges.

For example, a negative study duration will be flagged as invalid, while a missing sleep duration will be treated according to a documented missing-data policy rather than automatically being interpreted as zero.

The system will maintain clear rules for correcting, rejecting, or flagging problematic records.

7.3. ETL Pipeline

The project will implement an ETL pipeline consisting of three main stages.

Extract: Read raw personal activity data from CSV files or supported input forms.

Transform: Standardize formats, validate values, remove duplicates where appropriate, handle missing information, calculate derived fields, and prepare records for storage.

Load: Insert validated records into the target relational database and update analytical tables or views.

The pipeline will be designed to be repeatable and testable. Where practical, the system will also track processing status and record validation errors to help diagnose problems.

7.4. Centralized Data Storage

The platform will store personal activity data in a relational database.

The database will be designed to keep daily records organized and to support analytical queries without requiring repeated manual processing of the original files.

Possible entities include users, daily activity records, activity categories, and data import history. The exact database schema will be finalized during the design phase.

7.5. Personal Analytics Dashboard

The dashboard will display key indicators and visualizations, including:

- Average daily sleep duration.

- Total and average study hours.

- Weekly physical activity.

- Daily screen time.

- Task completion rate.

- Changes in the productivity score over time.

- Comparisons between weekdays and weekends.

- Trends across selected periods.

Users will be able to filter the displayed results by date range and compare selected indicators.

7.6. Behavioral Pattern Analysis

The platform will analyze relationships between recorded activity categories.

For example, it may compare sleep duration with the user's self-reported productivity score, examine study patterns across the week, or investigate whether screen time and study duration tend to vary together.

The analysis will distinguish between correlation and causation. A relationship between two metrics does not automatically mean that one causes the other.

7.7. What-if Scenario Simulation

The what-if module will allow users to explore hypothetical changes to selected aspects of their daily routine.

For example, a user may compare the following scenarios:

- The current recorded routine.

- A hypothetical routine with an additional hour of sleep.

- A routine with a different study schedule.

- A routine with reduced screen time.

- A routine with additional physical activity.

The system will calculate the differences between the selected scenario and the historical baseline.

In the initial version, these results will be based on explicit rules, averages, or simple statistical estimates. If the system lacks sufficient historical evidence, it will display the scenario as hypothetical rather than presenting an unsupported productivity prediction.

7.8. Basic Predictive Analytics

If enough consistent historical records are available, the project may implement a basic forecasting model to estimate selected future metrics, such as study duration or a self-reported productivity score.

The model will be evaluated using appropriate historical validation methods. Predictions will be labeled as estimates, and the system will communicate their limitations.

Predictive analytics will be treated as an extension of the core data engineering system rather than a requirement that takes priority over data quality and reliable processing.

8. Data Sources and Data Categories

The project will initially use synthetic sample data, manually entered records, or user-provided CSV files.

This approach allows the system to be developed and tested without depending on external services or collecting sensitive information from third parties.

The planned data categories are:

1. Sleep Data

- Date.

- Sleep duration in hours.

- Optional sleep quality rating.

2. Study Data

- Date.

- Study duration in hours.

- Number of study sessions.

- Subject or activity category, if applicable.

3. Physical Activity Data

- Date.

- Activity type.

- Activity duration in minutes.

4. Screen Time Data

- Date.

- Total screen time in hours.

- Optional category of screen use, if available.

5. Productivity Data

- Date.

- Number of planned tasks.

- Number of completed tasks.

- Optional self-reported productivity score.

These categories will be integrated using a common date field and a consistent data model.

The initial implementation will avoid collecting unnecessary identifiers or sensitive information. If multiple users are supported in a future version, appropriate access controls and separation of user records will be required.

9. Data Engineering Architecture

The proposed system will contain the following logical layers:

Layer 1: Data Sources

Raw data will originate from CSV files, manual entries, or other permitted sources.

Layer 2: Data Ingestion

The ingestion component will read incoming records, identify their source, and prepare them for processing.

Layer 3: ETL and Data Quality

A processing pipeline will clean, validate, standardize, and transform the incoming data.

Layer 4: Data Storage

The processed records will be stored in a relational database. Analytical views or summary tables may be added to support efficient reporting.

Layer 5: Analytics

The analytical layer will calculate metrics, aggregate daily and weekly results, examine trends, and prepare data for scenario comparisons.

Layer 6: Visualization

An interactive dashboard will present the analytical results through charts, tables, filters, and summary indicators.

Layer 7: Scenario Analysis

A what-if component will compare hypothetical inputs with the historical baseline and display differences using transparent calculation rules.

Data Flow

Raw Data → Data Ingestion → ETL Pipeline → Data Validation → Relational Database → Analytics → Dashboard and What-if Analysis

This architecture separates data processing, storage, analysis, and presentation so that each component can be tested and improved independently.

10. Proposed Technologies

The following technologies are proposed for the initial implementation. Final selections may be adjusted according to project requirements and available resources.

Python: For data processing, ETL development, validation, and analytical calculations.

Pandas: For reading CSV files, cleaning data, transforming records, and preparing analytical datasets.

SQL: For querying, aggregating, and analyzing structured data.

PostgreSQL or MySQL: For relational data storage. One database will be selected for the final implementation.

Streamlit: For developing an interactive dashboard using Python.

Plotly or a comparable visualization library: For creating interactive charts.

Git and GitHub: For version control, project documentation, and collaborative development.

Optional forecasting libraries: Statistical or machine learning libraries may be used if the project reaches a stage where forecasting is justified by the available data.

The project will prioritize a working and reliable data pipeline before adding unnecessary technologies.

11. Proposed Database Design

The database will be designed to support reliable storage, traceability, and analytical queries.

A preliminary design may include the following tables:

Users

Stores a minimal user identifier and any necessary non-sensitive configuration.

Daily_Activity

Stores the user's daily records, including sleep duration, study hours, screen time, physical activity, and selected productivity metrics.

Activity_Records

Optionally stores individual activities when more detailed event-level tracking is required.

Data_Import_Log

Records information about data import operations, including processing status, timestamps, and the number of accepted or rejected records.

Scenario_Results

Optionally stores saved hypothetical scenarios and their calculated outputs for later comparison.

The final schema will depend on whether the project stores one daily summary per user or multiple detailed activity records per day.

Database constraints and validation rules will be used to prevent invalid values and maintain consistency.

12. Data Warehouse and Analytical Model

The project may introduce a small analytical warehouse or reporting model to demonstrate data warehousing concepts.

A possible design is a star schema containing a central fact table and supporting dimension tables.

Fact_Daily_Activity: Stores measurable daily indicators, such as study duration, sleep duration, screen time, exercise duration, and completed tasks.

Dim_Date: Contains date-related attributes such as day, month, year, weekday, and weekend status.

Dim_User: Identifies the user when multi-user functionality is implemented.

Dim_Activity: Describes activity categories if detailed activity records are included.

This structure can support queries such as weekly study totals, monthly averages, and comparisons across activity categories.

The project will not require a complex enterprise-scale warehouse. A small, clearly documented analytical model is sufficient to demonstrate dimensional modeling and aggregation.

13. Analytical Metrics

The platform will calculate several indicators from the collected data.

Average Study Duration

The average number of study hours recorded per day over a selected period.

Average Sleep Duration

The average recorded sleep duration over a selected period.

Task Completion Rate

Completed tasks divided by planned tasks, multiplied by 100, when the number of planned tasks is greater than zero.

Daily Screen Time

The total recorded screen time for a selected day.

Weekly Activity Summary

The sum or average of selected activity metrics across a calendar week.

Routine Consistency

A descriptive measure of how consistently a selected habit occurs across recorded days. The calculation method will be documented in the implementation.

Productivity Trend

The change in the user's self-reported productivity score over time, where such scores have been collected.

All metrics will be calculated from available and validated records. Missing or incomplete data will be disclosed so that users can interpret the results correctly.

14. What-if Analysis Methodology

The what-if analysis component will compare an existing historical baseline with one or more hypothetical scenarios.

The process will involve the following steps:

1. Select a historical date range to establish a baseline.

2. Calculate the relevant baseline metrics.

3. Allow the user to modify selected hypothetical inputs.

4. Apply a documented calculation method to the new scenario.

5. Compare the scenario output with the baseline.

6. Display the differences and explain the assumptions used.

For example, the platform may calculate the average recorded study duration over the previous two weeks and compare it with a hypothetical weekly schedule containing additional planned study time.

A simple scenario may calculate planned time directly. A more advanced scenario may estimate an outcome using a statistical relationship learned from historical data, if the data is adequate.

The system will not assume that adding one hour of study or sleep necessarily produces a specific increase in productivity. Any estimated effect must be supported by a defined model and presented with appropriate limitations.

15. Privacy and Ethical Considerations

Personal routine data can reveal information about an individual's lifestyle. Therefore, privacy will be considered throughout the design and implementation of the project.

The initial implementation will use synthetic, manually entered, or user-provided data. It will not require access to private messages, personal accounts, or device-level monitoring.

The system will collect only the information required for the intended analytical functions.

If personal records are used, the user should understand what is stored and how it is processed. Access to records should be restricted appropriately, and secrets such as database passwords must not be uploaded to the public GitHub repository.

The project will avoid making medical diagnoses or presenting lifestyle correlations as medical conclusions.

The digital twin will be treated as an analytical aid rather than an authoritative representation of the individual. Its results will depend on the quality, completeness, and relevance of the recorded data.

16. Project Scope

In Scope

- Importing activity data from CSV files or supported manual input.

- Cleaning and validating incoming records.

- Developing a repeatable ETL pipeline.

- Designing a relational database.

- Building basic analytical queries and metrics.

- Developing an interactive dashboard.

- Analyzing historical trends and relationships.

- Implementing transparent what-if comparisons.

- Documenting data quality, assumptions, and limitations.

- Using GitHub for version control and project documentation.

Out of Scope for the Initial Version

- Continuous monitoring of all device activity.

- Automatic access to private social media or personal accounts.

- Medical diagnosis or treatment recommendations.

- Guaranteed prediction of human behavior.

- A full biological or real-time physical simulation of a human being.

- Large-scale production infrastructure or enterprise cloud deployment.

These limitations keep the project realistic and achievable while preserving its core data engineering value.

17. Expected Results and Deliverables

At the end of the project, the expected deliverables are:

1. A documented GitHub repository containing the project files and source code.

2. A project proposal describing the problem, objectives, architecture, and planned implementation.

3. A sample dataset containing structured daily activity records.

4. An ETL pipeline for extracting, cleaning, transforming, and loading data.

5. A relational database schema and SQL queries.

6. An analytical model or reporting views for daily and weekly metrics.

7. An interactive dashboard showing activity patterns and trends.

8. A what-if analysis component for comparing hypothetical routines.

9. Basic forecasting functionality if sufficient data and development time are available.

10. Documentation explaining how to install, configure, run, and test the system.

11. A final report describing the implementation, testing, results, limitations, and possible future improvements.

The project will be considered successful when it demonstrates a repeatable end-to-end data workflow and produces reliable, interpretable analytical outputs from the available data.

18. Testing and Evaluation

The project will be evaluated through functional and data quality tests.

Data Quality Testing: Verify that invalid values, duplicate records, and missing fields are handled according to documented rules.

ETL Testing: Confirm that valid records are transformed correctly and loaded into the database.

Database Testing: Verify constraints, relationships, and the correctness of analytical queries.

Metric Testing: Compare calculated indicators with manually verified examples.

Dashboard Testing: Confirm that charts, date filters, and summary indicators reflect the underlying data.

Scenario Testing: Verify that what-if calculations follow the documented formulas and that outputs are labeled appropriately.

Performance Testing: Measure processing time for representative sample datasets and check that analytical queries complete within reasonable time.

The evaluation will focus on correctness, repeatability, data integrity, usability, and the transparency of analytical assumptions.

19. Project Plan

The project will be implemented in stages.

Phase 1: Requirements and Proposal

Finalize the problem statement, objectives, project scope, data categories, and initial architecture.

Phase 2: Data Design and Sample Dataset

Define the data dictionary, prepare sample records, and design the initial database schema.

Phase 3: ETL Development

Implement extraction, transformation, validation, and loading procedures.

Phase 4: Database and Analytical Queries

Create database tables, constraints, summary queries, and analytical views.

Phase 5: Dashboard Development

Build the dashboard and visualize key personal activity metrics.

Phase 6: What-if Analysis

Implement scenario comparisons using transparent rules or appropriate statistical methods.

Phase 7: Testing and Documentation

Test the complete pipeline, document the system, and prepare the final project report and demonstration.

The duration of each phase will be determined according to the academic deadline and the final requirements.

20. Future Enhancements

Possible future improvements include:

- Integration with permitted wearable-device exports.

- Automated scheduled data ingestion.

- More advanced forecasting models.

- Personalized trend summaries.

- Comparison of multiple routine scenarios.

- Improved data quality monitoring.

- Secure multi-user functionality.

- Cloud deployment and automated pipeline scheduling.

These enhancements will be considered only after the core system has been implemented and evaluated.

21. Conclusion

The Personal Digital Twin project proposes a practical application of data engineering to personal behavior analytics. It transforms daily activity records into a structured, integrated, and analyzable dataset through data ingestion, ETL processing, relational storage, and analytical modeling.

The platform will help users explore historical trends, understand relationships between recorded habits, and compare hypothetical routine changes through an interactive dashboard.

The project's main contribution is not simply the visualization of personal statistics. It is the design and implementation of a complete data pipeline that addresses data quality, integration, storage, analytical processing, and reproducibility.

By combining these components, the project will demonstrate how data engineering can convert raw personal data into meaningful analytical information while maintaining clear assumptions, appropriate privacy practices, and realistic expectations about prediction.

The resulting system will provide a foundation for future enhancements in personalized analytics and data-driven decision support.